

Problem 25.11

Show that

$$a_{\overline{n}|} < a_{\overline{n}|}^{(m)} < \bar{a}_{\overline{n}|} < \ddot{a}_{\overline{n}|}^{(m)} < \ddot{a}_{\overline{n}|}.$$

Hint: See Example 10.15.

Solution

For $m > 1$, the corresponding rates satisfy the ordering

$$d < d^{(m)} < \delta < i^{(m)} < i.$$

Taking reciprocals reverses the inequalities:

$$\frac{1}{d} > \frac{1}{d^{(m)}} > \frac{1}{\delta} > \frac{1}{i^{(m)}} > \frac{1}{i}.$$

The present value factors for annuities over n years can be written as

$$a_{\overline{n}|} = \frac{1 - v^n}{i},$$

$$a_{\overline{n}|}^{(m)} = \frac{1 - v^n}{i^{(m)}},$$

$$\bar{a}_{\overline{n}|} = \frac{1 - v^n}{\delta},$$

$$\ddot{a}_{\overline{n}|}^{(m)} = \frac{1 - v^n}{d^{(m)}},$$

and

$$\ddot{a}_{\overline{n}|} = \frac{1 - v^n}{d}.$$

Since

$$1 - v^n > 0,$$

we may multiply the reciprocal inequality by $1 - v^n$ without changing the order:

$$\frac{1 - v^n}{d} > \frac{1 - v^n}{d^{(m)}} > \frac{1 - v^n}{\delta} > \frac{1 - v^n}{i^{(m)}} > \frac{1 - v^n}{i}.$$

Reversing this chain gives

$$\frac{1 - v^n}{i} < \frac{1 - v^n}{i^{(m)}} < \frac{1 - v^n}{\delta} < \frac{1 - v^n}{d^{(m)}} < \frac{1 - v^n}{d}.$$

Substituting the annuity symbols,

$$a_{\overline{n}|} < a_{\overline{n}|}^{(m)} < \bar{a}_{\overline{n}|} < \ddot{a}_{\overline{n}|}^{(m)} < \ddot{a}_{\overline{n}|}.$$

Thus,

$$\boxed{a_{\overline{n}|} < a_{\overline{n}|}^{(m)} < \bar{a}_{\overline{n}|} < \ddot{a}_{\overline{n}|}^{(m)} < \ddot{a}_{\overline{n}|}}.$$